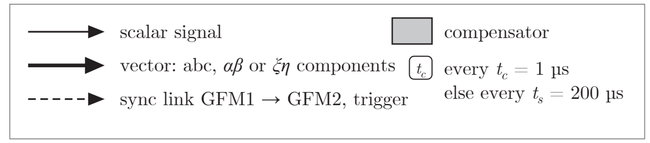
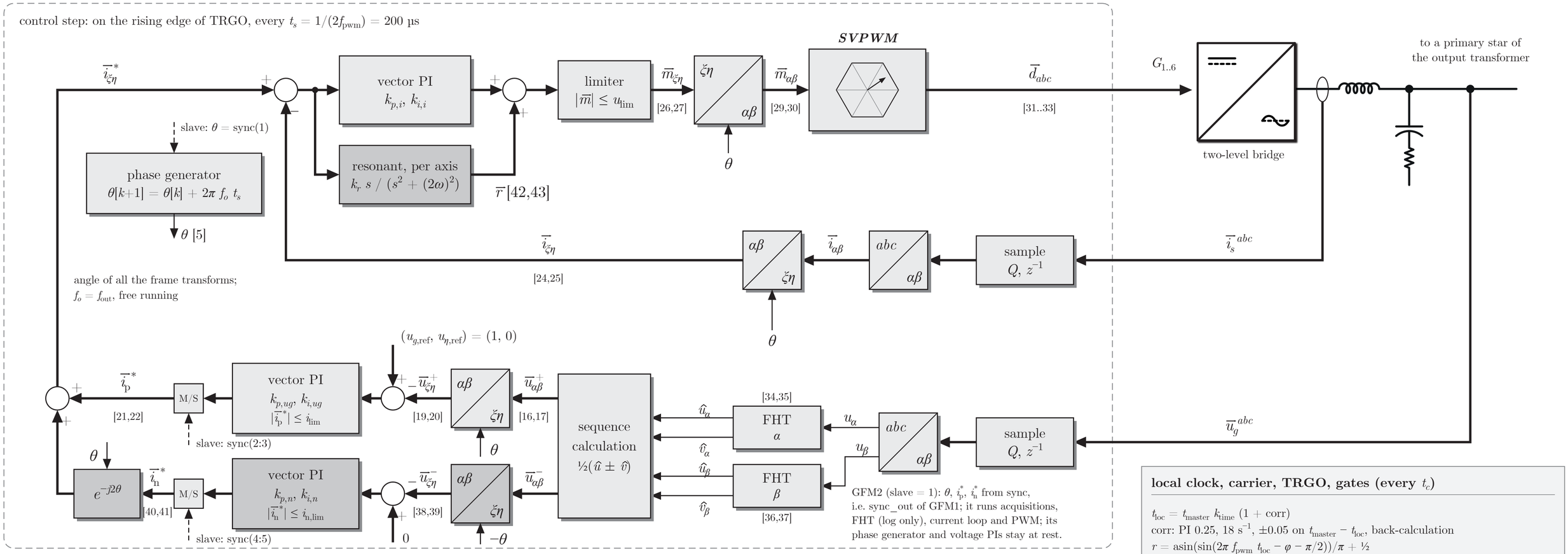
Grid-forming converter control, VFC0 test bench

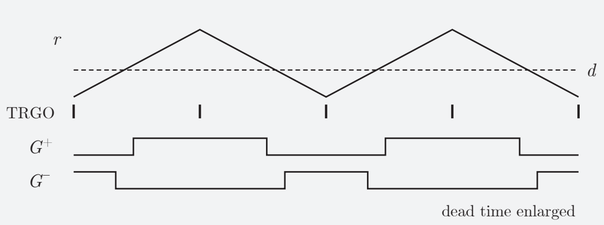
afe_dq_ctrl_pwm: MATLAB Function of the library block grid_forming_dqctrl (dev 57cdc00), as used by the VFC0 model; GFM1 master, GFM2 slave;
negative-sequence compensator on (nsc_enable = 1). Signal names as in the code, [n] = column of ctrl_log (grid_forming_log_index).
Per unit: $u_{\mathrm{bez}} = 326.6$ V (phase peak), $i_{\mathrm{bez}} = 777.8$ A; voltages and currents of the control in pu.





local clock, carrier, TRGO, gates (every t_c)

$t_{\mathrm{loc}} = t_{\mathrm{master}}\, k_{\mathrm{time}}\, (1 + \mathrm{corr})$
corr: PI 0.25, 18 s^{-1}, ±0.05 on $t_{\mathrm{master}} - t_{\mathrm{loc}}$, back-calculation
$r = \mathrm{asin}(\sin(2\pi\, f_{\mathrm{pwm}}\, t_{\mathrm{loc}} - \varphi - \pi/2))/\pi + ½$
TRGO: $r \le 2 f_{\mathrm{pwm}} t_c$ or $r \ge 1 - 2 f_{\mathrm{pwm}} t_c$ (top and bottom)
control step on its rising edge: $t_s = 1/(2f_{\mathrm{pwm}}) = 200$ µs
gates: upper on if $r \ge d + t_{\mathrm{dead}} f_{\mathrm{pwm}}$, lower on if $r \le d - t_{\mathrm{dead}} f_{\mathrm{pwm}}$
d in [0.02, 0.98], $t_{\mathrm{dead}} = 3$ µs, gate +15 / −10 V



FHT, first harmonic tracker

per axis (α, β), resonant observer:
$A = [0, 1; -\omega_0^2, -\delta\omega_0]$, $C = [1, 0]$
$\omega_0 = 2\pi\, f_o$, $\delta = 0.05$
poles of $A - LC$: $(-1, -4)\,\omega_0$

discrete (C code), $a = -\omega_0^2$, $b = -\delta\omega_0$:
$a_{11} = 1 + ½\, a\, t_s^2$, $a_{12} = t_s + ½\, b\, t_s^2$
$a_{21} = a\, t_s + ½\, ab\, t_s^2$
$a_{22} = 1 + b\, t_s + ½\, (a + b^2)\, t_s^2$
$e[k] = u[k] - \hat{u}[k-1]$
$x_1 \leftarrow a_{11}\, x_1 + a_{12}\, x_2 + l_1\, e$
$x_2 \leftarrow a_{21}\, x_1 + a_{22}\, x_2 + l_2\, e$
$\hat{u}[k] = x_1$ (before the update)
$\hat{v}[k] = x_2/\omega_0$ (after the update)

$\hat{v}$: $\hat{u}$ shifted by 90° (lead) at f_o
reset (states 0) while $t < 12.5$ ms

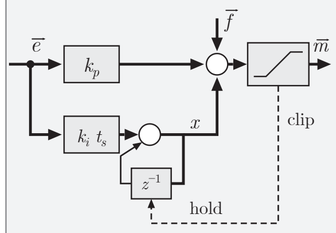
sequences, frames, phase

$u_\alpha^+ = ½\, (\hat{u}_\alpha + \hat{v}_\beta)$
$u_\beta^+ = ½\, (\hat{u}_\beta - \hat{v}_\alpha)$
$u_\alpha^- = ½\, (\hat{u}_\alpha - \hat{v}_\beta)$
$u_\beta^- = ½\, (\hat{u}_\beta + \hat{v}_\alpha)$

positive frame (θ):
$u_\xi = u_\alpha^+ \cos\theta + u_\beta^+ \sin\theta$
$u_\eta = -u_\alpha^+ \sin\theta + u_\beta^+ \cos\theta$
negative frame ($-\theta$):
$u_\xi^- = u_\alpha^- \cos\theta - u_\beta^- \sin\theta$
$u_\eta^- = u_\alpha^- \sin\theta + u_\beta^- \cos\theta$
rotation $e^{-\hat{\jmath}2\theta}$ into the positive frame:
$i^*_\xi = i^*_{\xi\mathrm{p}} + c_2\, i^*_{\xi\mathrm{n}} + s_2\, i^*_{\eta\mathrm{n}}$
$i^*_\eta = i^*_{\eta\mathrm{p}} - s_2\, i^*_{\xi\mathrm{n}} + c_2\, i^*_{\eta\mathrm{n}}$
$c_2 = \cos 2\theta$, $s_2 = \sin 2\theta$

A negative sequence at f_o is constant
in the negative frame: the PIs see DC
values; no average or notch filter.

vector PI with limiter



$m = k_p\, e + x + f$
normal: $x \leftarrow x + k_i\, t_s\, e$
clip: $x \leftarrow (1 - c_{\mathrm{rel}}\, t_s)\, x$ (held)
$|m| = \sqrt{(m_\xi^2 + m_\eta^2)}$
$m \leftarrow m \cdot lim/|m|$ if $|m| \ge lim$
clip = ($|m| \ge thr$), used next step

voltage PI, compensator:
$lim = thr = 1.65$ pu
current loop: $lim = thr = u_{\mathrm{lim}} = 1$
f: resonant terms in the current loop,
0 in the voltage PIs

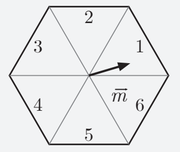
resonant term at $2f_o$

$R(s) = k_r\, s/(s^2 + \omega_r^2)$
$\omega_r = 2 \cdot 2\pi\, f_o = 209.4$ rad/s
$k_r = 2\, k_{i,i} = 70\ \mathrm{s}^{-1}$

discrete, $a = 2\sin(\omega_r\, t_s/2)$:
$x_1 \leftarrow x_1 + t_s\, k_r\, e - a\, x_2$
$x_2 \leftarrow x_2 + a\, x_1$ (new x_1)
$r = x_1$, added to the PI output
eigenvalues on the unit circle
at $\pm\omega_r\, t_s$: resonance exactly at ω_r
clip: input off, states × $(1 - c_{\mathrm{rel}} t_s)$

Why: $i^*_{\mathrm{n}}\, e^{-\mathrm{j}2\theta}$ is a sinusoid at $2f_o$
in the positive frame; the PI alone
would track it with an error. The PI
tracks the DC part (positive sequence),
the resonant term the negative
sequence.

SVPWM, acquisitions



inputs $\overline{m}_{\alpha\beta} \cdot \sqrt{3}/2$, $k = (u_y + 1)/2 = ½$
sector 1:
$t_1 = m_\alpha - m_\beta/\sqrt{3}$, $t_2 = 2m_\beta/\sqrt{3}$
$t_0 = 1 - t_1 - t_2$
$d_a = k\, t_0$, $d_b = d_a + t_1$, $d_c = d_b + t_2$
upper switch on while $r \ge d$

acquisitions on TRGO:
$x_q = q \cdot \mathrm{round}(x/q)$, $q = 2^{-11}$ pu
one step delay: the step uses the
samples of the previous TRGO;
i_g with gain −1

parameters (VFC0 model, GFM1 and GFM2)

f_o	16.67 Hz	ω_0, δ	104.7 rad/s, 0.05
f_{pwm}	2.5 kHz, double update	l_1, l_2	0.100, 5.61
t_s, t_c	200 µs, 1 µs	c_{rel}	0.001 s^{-1}
$k_{p,ug}$, $k_{i,ug}$	0.5, 18 s^{-1}	q	2^{-11} pu
i_{lim}	1.65 pu	t_{dead}	3 µs
$k_{p,n}$, $k_{i,n}$	0.5, 18 s^{-1}	$u_{g,\mathrm{ref}}$	1 pu (400 V)
$i_{\mathrm{n,lim}}$	1.65 pu	u_{bez}	326.6 V
$k_{p,i}$, $k_{i,i}$	0.4, 35 s^{-1}	i_{bez}	777.8 A
k_r	70 s^{-1}	slave	0 (GFM1), 1 (GFM2)
u_{lim}	1		